Evan Gibbs

Dover, Pennsylvania evangibbs717@gmail.com 717-817-1843

evan-gibbs717 EvanGibbs717

Education

York College of Pennsylvania *Aug 2022 – Dec 2025*

BS in Mathematics (Minors in Biology and Data Science)

- GPA: 4.0/4.0
- **Important Coursework:** Real Analysis I-II, Linear Algebra I-II, Abstract Algebra I, Topology, Graph Theory and Combinatorics, Modern Geometry, Data Mining and Machine Learning, Probability Theory and Statistics, ODE(s), Calculus I-III, Data Analysis I, Biostatistics.

Thesis: The Representation of Positive Integers by Positive, Definite Quadratic Forms (Advised by Dr.Haiping Yuan)

Technological and Programming Skills

Languages: MATLAB, Python, LaTeX, R.

Technologies: Git(hub), MATHCAD, GraphPad Prism, Microsoft Office (Certified).

Research Experience

Quantitative Research in the Life and Social Sciences Program REU Student *Tempe, AZ*

Arizona State University, Simon Levin MCMSC *May 2025 – July 2025*

- Developing a mathematical model of *Wolbachia*-dengue dynamics, where optimal control theory is used to evaluate release strategies of *Wolbachia*-infected mosquitoes for disease suppression.
- Performing a classical, mathematical analysis of a system of six nonlinear ODEs, while computing optimal controls with GPOPS-II software.
- Comparing the trade-offs between intervention costs and infection reduction with the aim of determining the most cost-effective strategy for reduction of dengue in local communities.

Student Researcher in Geometric Data Analysis and Manifold Learning *Chicago, IL*

Work affilated with NITMB, Northwestern University and University of Chicago *June 2024 – Dec 2025*

- Using manifold learning techniques to learn the state-space of *D. melanogaster* wing morphology.
- Leveraging Wasserstein geometry to provide new insights to quantifying morphological variation among various wing strain, sex, and characteristic combinations.
- Capitalizing on the manifold structure induced by the Wasserstein metric to determine the optimal transport maps between wings, as well as gauge the intrinsic dimensionality of the wing-space at given points.

National Institute of Theory and Mathematics in Biology SURP Student *Chicago, IL*

NITMB (Northwestern University and University of Chicago) *June 2024 – Aug 2024*

- Used manifold learning techniques to learn the state-space of *D. melanogaster* wing morphology.
- Performed techniques such as PCA, UMAP, and tSNE to understand modes of variation in *D. melanogaster*.
- Began preliminary analysis on *D. melanogaster* wing morphology in Wasserstein space.

Student Researcher in Aquaponics Modeling *York, PA*

York College of Pennsylvania Department of Biology *Jun 2023 – May 2024*

- Constructed a system of recursive, piecewise equations to model nitrate concentration across a decoupled aquaponics system.
- Programmed an adaptable simulation of a decoupled system in MATLAB to predict nitrate concentrations based on user-specific aquaponics systems.
- Validated this mathematical system against 6 decoupled aquaponics systems consisting of 108 tilapia and 36 kale plants to determine the efficacy and improvements for the model.

Research Assistant in Aquaponics Sustainability *York, PA*
York College of Pennsylvania Department of Biology *Jan 2023 – May 2023*

- Created a feed for Blue Tilapia consisting of cafeteria food waste to present a sustainable alternative to conventional aquaculture feeds.
- Raised 216 tilapia on varying diets of feed to show that a food waste alternative performs as well as fish-meal diets.

Research Presentations

Tangent Talks Lecture Series - York, PA Nov 6 2025

Gibbs, E, Elhassan, K, Sanchez, J, Gohil, U, Arias Castro, H, Cevallos Chavez, J. "Optimal Control Approach to Developing Sex-Structured Release Strategies for Dengue Prevention," York College of Pennsylvania.

MAA EPADEL Section Meeting - York, PA Oct 4 2025

Gibbs, E, Elhassan, K, Sanchez, J, Gohil, U, Arias Castro, H, Cevallos Chavez, J. "Optimal Control Approach to Developing Sex-Structured Release Strategies for Dengue Prevention," MAA EPADEL.

QRLSSP Symposium - Tempe, AZ Jul 17 2025

Gibbs, E, Elhassan, K, Sanchez, J, Gohil, U, Arias Castro, H, Cevallos Chavez, J. "Optimal Control Approach to Developing Sex-Structured Release Strategies for Dengue Prevention," Arizona State University.

York College Undergraduate Research Showcase - York, PA Apr 30 2025

Gibbs, E, Riesenfeld, S, Weinberger, S, Carthew, R, Robinett, R. "Elucidating the Heterogeneity of Wing Morphology in *D.melanogaster* using Wasserstein Geometry," York College of Pennsylvania.

100th Annual PA Academy of Sciences Symposium - Harrisburg, PA Apr 12 2025

Gibbs, E, Riesenfeld, S, Weinberger, S, Carthew, R, Robinett, R. "Elucidating the Heterogeneity of Wing Morphology in *D.melanogaster* using Wasserstein Geometry," PA Academy of Sciences.

MAA EPADEL Section Meeting - Mechanicsburg, PA Mar 29 2025

Gibbs, E, Riesenfeld, S, Weinberger, S, Carthew, R, Robinett, R. "Elucidating the Heterogeneity of Wing Morphology in *D.melanogaster* using Wasserstein Geometry," MAA EPADEL.

NITMB SURP Summer Symposium - Chicago, IL Aug 9 2024

Gibbs, E, Riesenfeld, S, Weinberger, S, Carthew, R, Robinett, R. "Learning the State-Space of *D.melanogaster* Wing Morphology Using Manifold Learning Techniques," NITMB.

99th Annual PA Academy of Sciences Symposium - Mechanicsburg, PA Apr 14 2024

Gibbs, E. "A Recursive MATLAB Model for Nitrate in a Decoupled Aquaponics System," PA Academy of Sciences.

York College Biology Department Research Symposium - York, PA Dec 4 2023

Gibbs, E. "A Recursive MATLAB Model for Nitrate in a Decoupled Aquaponics System," York College of Pennsylvania.

York College Biology Department Research Symposium - York, PA May 3 2023

Gibbs, E. "Food Waste Diet for Blue Tilapia (*Oreochromis aureus*) Utilizing Cafeteria Proteins," York College of Pennsylvania.

Conferences and Workshops Attended

SACNAS NDiSTEM 2025	Oct 30 - Nov 1 2025
2025 Fall Section Meeting of MAA EPADEL	Oct 4 2025
NITMB MathBio Convergence Conference	Aug 11-15 2025
100thAnnual Meeting of the PA Academy of Sciences, Harrisburg University of Science and Technology	Apr 11-13 2025
2025 Spring Section Meeting of MAA EPADEL	Mar 29 2025
NITMB Annual Emerging Directions Workshop	Feb 17-21 2025
99thAnnual Meeting of the PA Academy of Sciences, Messiah University	April 12-14 2024

Academic Awards

York College of Pennsylvania Mathematics Departmental Honors	Dec 17 2025
2nd Place Oral Presentation Award at York College of Pennsylvania Undergraduate Research Showcase	Apr 30 2025
2nd Place Spychala Oral Presentation Award at 100th Annual Meeting of the PA Academy of Sciences, Harrisburg University of Science and Technology	Apr 13 2025
1st Place Spychala Poster Presentation Award at 99th Annual Meeting of the PA Academy of Sciences, Messiah University.	Apr 14 2024
Alpha Chi National Honor Society (Top 5% of Junior Class)	Apr 26 2024
York College of Pennsylvania Dean's List Fall 2025	Dec 15 2025
York College of Pennsylvania Dean's List Spring 2025	May 20 2025
York College of Pennsylvania Dean's List Fall 2024	Dec 15 2024
York College of Pennsylvania Dean's List Spring 2024	May 20 2024
York College of Pennsylvania Dean's List Fall 2023	Dec 15 2023
York College of Pennsylvania Dean's List Spring 2023	May 20 2023
York College of Pennsylvania Dean's List Fall 2022	Dec 15 2022

Funding Awards

York College of Pennsylvania Undergraduate Research Grant	Sept 2025
Travel Funding for the NITMB MathBio Convergence Conference	Aug 2025
Travel Funding for the NITMB Emerging Directions Conference	Feb 2025

Related Experience

Mathematics Teaching Assistant *York, PA*
York College of Pennsylvania Academic Support Center *Aug 2023 – Dec 2025*

- Host review sessions for students in Calculus I-II and Ordinary Differential Equations.
- Lecture on review problems prior to course examinations.

Mathematics, Statistics, and Physics Tutor *York, PA*
York College of Pennsylvania Academic Support Center *Jan 2023 – Dec 2025*

- Tutor students in Calculus I-III, Linear Algebra I, Ordinary Differential Equations, Probability and Statistics, Biostatistics, Engineering Physics.

Economics and Student Managed Fund Researcher *York, PA*
York College of Pennsylvania Graham School of Business *Dec 2024 – Sept 2025*

- Designing a web-based application for screening stocks based on various economic metrics.
- Implementing Artificial Intelligence and Machine Learning techniques for stock buying decisions.
- Automating the purchasing, selling, and holding pipeline for York College's Student Managed Fund.

Laboratory Assistant and SDS Chemical Inventory *York, PA*
York College of Pennsylvania Department of Biology *May 2023 – Dec 2023*

- Lab assisted for an introductory molecular biology class.
- Completed a comprehensive and updated SDS chemical inventory for the York College Biology Department.

Extra-Cirriculars

York College of Pennsylvania Mathematics Club - President *York, PA*
York College of Pennsylvania Mathematics Department *Nov 2024 – Dec 2025*

York Men's Club Soccer - President *York, PA*
York College of Pennsylvania *Jan 2023 – Dec 2025*